Topology & Multivariable Calculus

FMA_2F002_EP

Lecture Notes

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These are personal lecture notes and may contain errors. For the official course material consult the lecturer.

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1 Norms in Vector Spaces

1.1 Setting

Let $E, F, G, \ldots$ be vector spaces over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$.

- If $\mathbb{K} = \mathbb{R}$, for $x \in \mathbb{R}$, $|x| = \max(x, -x)$.
- If $\mathbb{K} = \mathbb{C}$, for $z = x + iy$, we have $|z| = \sqrt{x^2 + y^2} = l$. In polar form, $z = le^{i\theta}$ where $l \geq 0$ and $\theta \in [0, 2\pi)$.

1.2 Norms

Definition 1: Norm

A norm on E is a map $N : E \to \mathbb{R}_{\geq 0} = \mathbb{R}^+ = [0, +\infty)$ satisfying the following 3 properties:

1. **Triangular inequality:** $\forall x, y \in E$, $N(x + y) \leq N(x) + N(y)$.
2. **Absolute homogeneity:** $\forall \lambda \in \mathbb{K}, \forall x \in E$, $N(\lambda x) = |\lambda| N(x)$.
3. **Positive definiteness:** $N(x) = 0 \Leftrightarrow x = 0$.

Remark 2

Positive definiteness is equivalent to the fact that N "separates points". That is, $\forall x, y \in E$, $N(x - y) = 0$ if and only if $x = y$.

Proposition 3: Reverse Triangular Inequality

The triangular inequality is equivalent to the reverse triangular inequality: $\forall x, y \in E$, $|N(x) - N(y)| \leq N(x - y)$.

Proof. Let $x, y \in E$. Then $N(x) = N(x - y + y) \leq N(x - y) + N(y)$, which implies $N(x) - N(y) \leq N(x - y)$. Similarly, $N(y) - N(x) \leq N(y - x)$. Since $N(x - y) = |-1| N(y - x)$, we obtain $|N(x) - N(y)| \leq N(x - y)$. □

Example 4: Common Norms

- $|\cdot|$ is a norm on $\mathbb{R}$.
- On $E = \mathbb{R}^d$, Sup Norm $||x||_\infty = \max_{i=1}^{d} |x_i|$.
- Euclidean norm on $\mathbb{R}^d$: $||x||_2 = \sqrt{\sum_{i=1}^{d} x_i^2}$.
- On $F = C^0([0, 1], \mathbb{R})$, Sup Norm $||f||_\infty = \sup_{x \in [0,1]} |f(x)|$.
- On F, L_1 norm $||f||_{L^1} = \int_0^1 |f(x)| dx$.
- On F, L_2 norm $||f||_{L^2} = \sqrt{\int_0^1 |f(x)|^2 dx}$.

Lemma 5: Norms on $\mathbb{R}$

Let $N : \mathbb{R} \to \mathbb{R}_{\geq 0}$ be a norm. Then $\exists \alpha > 0$ such that $N(\cdot) = \alpha|\cdot|$.

PROOF. Let $x \in \mathbb{R}$. Then $N(x) = N(x \cdot 1) = |x|N(1)$. Let $\alpha = N(1) > 0$. □

Lemma 6: Euclidean Norm on $\mathbb{R}^d$

$||\cdot||_2$ is a norm on $E = \mathbb{R}^d$.

PROOF. Absolute homogeneity is obvious. For positive definiteness: $||x||_2 = 0 \Rightarrow \sum_{i=1}^d x_i^2 = 0 \Rightarrow x_i = 0$ for all $i = 1, \ldots, d \Rightarrow x = 0_E$. For the triangular inequality, we use the Euclidean scalar product $\langle x, y\rangle = \sum_{i=1}^d x_i y_i$. This product is symmetric, $\mathbb{R}$-bilinear, and $\langle x, x\rangle = 0 \Rightarrow x = 0_E$. We have $||x + y||_2 = \sqrt{\langle x + y, x + y\rangle} = \sqrt{||x||_2^2 + ||y||_2^2 + 2\langle x, y\rangle}$. By the Cauchy-Schwarz inequality, $|\langle x, y\rangle| \leq ||x||_2||y||_2$. Thus, $||x + y||_2 \leq \sqrt{||x||_2^2 + 2||x||_2||y||_2 + ||y||_2^2} = ||x||_2 + ||y||_2$. □

Lemma 7: Sup Norm on $C^0([0, 1], \mathbb{R})$

$||\cdot||_\infty$ is a norm on $F = C^0([0, 1], \mathbb{R})$.

PROOF. For any $f \in F$, f is bounded. Positive homogeneity: $||\lambda f||_\infty = \sup|\lambda f(x)| = |\lambda|||f||_\infty$. Positive definiteness: If $||f||_\infty = 0$, then $0 \leq |f(x)| \leq \sup|f(x)| = 0$, which implies $f(x) = 0$ for all x, so $f = 0_F$. Triangular inequality: $|(f + g)(x)| = |f(x) + g(x)| \leq |f(x)| + |g(x)| \leq ||f||_\infty + ||g||_\infty$. Taking the supremum yields the result. □

1.3 Distances

Definition 8: Distance

A distance on a set $X \neq \emptyset$ is a map $d : X^2 \to \mathbb{R}^+$ satisfying:

1. **Triangular inequality:** $\forall x, y, z \in X$, $d(x, z) \leq d(x, y) + d(y, z)$.
2. **Symmetry:** $\forall x, y \in X$, $d(x, y) = d(y, x)$.
3. **Positive definiteness:** $\forall x, y \in X$, $d(x, y) = 0 \Leftrightarrow x = y$.

Example 9: Discrete Distance

Let $X \neq \emptyset$. The discrete distance is defined as $d(x, y) = 0$ if $x = y$, and $d(x, y) = 1$ if $x \neq y$.

Proposition 10: Distance Induced by a Norm

Let (E, N) be a normed vector space (NVS) over $\mathbb{K}$. The map $d_N : E^2 \to \mathbb{R}^+$ given by $d(x, y) = N(x - y)$ is a distance on E.

PROOF. Symmetry: $d(x, y) = N(x - y) = N(y - x) = d(y, x)$. Positive definiteness: $d(x, y) = 0 \Leftrightarrow N(x - y) = 0 \Rightarrow x - y = 0 \Rightarrow x = y$. Triangular inequality: $d(x, z) = N(x - z) =$

$N(x-y+y-z) \leq N(x-y)+N(y-z) = d(x,y)+d(y,z)$. □

1.4 Balls and Convexity

Definition 11: Balls

Let (X,d) be a metric space. For $x_0 \in X$ and $r \in \mathbb{R}^+$, we define:

- **Open ball:** $B(x_0,r) = \{x \in X \mid d(x_0,x) < r\}$.
- **Closed ball:** $\overline{B}(x_0,r) = \{x \in X \mid d(x_0,x) \leq r\}$.

In E, $B(0,1)$ is defined as the unit ball.

Definition 12: Segments and Convexity

For $x,y \in E$, the closed segment is $[x,y] = \{tx+(1-t)y \mid t \in [0,1]\}$. The open segment is $(x,y) = [x,y] \setminus \{x,y\}$. A subset $C \subseteq E$ is **convex** if for all $x,y \in C$, the segment $[x,y] \subseteq C$.

Proposition 13: Convex Subsets

1. Convex subsets of $\mathbb{R}$ are intervals.
2. Balls in a NVS are convex.

Proof. We prove (2) for $B = B(x_0,r)$ with $r > 0$. Let $x,y \in B$, meaning $N(x-x_0) < r$ and $N(y-x_0) < r$. Let $z = tx+(1-t)y \in [x,y]$ with $t \in [0,1]$. We have $d(z,x_0) = N(z-x_0) = N(t(x-x_0)+(1-t)(y-x_0))$. By the triangular inequality, $d(z,x_0) \leq tN(x-x_0)+(1-t)N(y-x_0) < tr+(1-t)r = r$. Thus, $z \in B$, which implies $[x,y] \subseteq B$, meaning B is convex. □

2 Sequences

Definition 1: Convergence of Sequences

Let $\underline{x} = (x_n)_{n\in\mathbb{N}}$ be a sequence in a $\mathbb{K}$-NVS E. We say $x_n \to x_\infty$ in E (wrt N) if $N(x_n - x_\infty) \to 0$ in $\mathbb{R}$. We denote this as $x_n \to x_\infty$ or $\lim x_n = x_\infty$. If a sequence does not converge, it diverges.

Example 2: Convergence in Different Norms

Consider $E = C^0([0,1],\mathbb{R})$ and the sequence of functions $f_n(x) = x^n$.

- For the sup norm $||\cdot||_\infty$, we have $||f_n||_\infty = 1$ for all n, so it does not converge to 0_E.
- For the L^1 norm, we have $||f_n||_{L^1} = \int_0^1 x^n dx = \frac{1}{n+1} \to 0$. Thus $f_n \to 0_E$ for $||\cdot||_{L^1}$.

Proposition 3: Uniqueness of the Limit

The limit of a convergent sequence is unique.

PROOF. Let $(x_n)_{n\in\mathbb{N}}$ be a sequence in E converging to both $x_\infty \in E$ and $x'_\infty \in E$. By the triangular inequality, $0 \leq N(x_\infty - x'_\infty) \leq N(x_\infty - x_n) + N(x_n - x'_\infty)$. Since $x_n \to x_\infty$ and $x_n \to x'_\infty$, both $N(x_\infty - x_n) \to 0$ and $N(x_n - x'_\infty) \to 0$ as $n \to \infty$. By the squeeze theorem, $N(x_\infty - x'_\infty) = 0$, which implies $x_\infty = x'_\infty$. □

Proposition 4: Continuity of the Norm

1. If $x_n \to x_\infty$ in E, then $N(x_n) \to N(x_\infty)$ in $\mathbb{R}$.
2. Hence, $(N(x_n))_{n\in\mathbb{N}}$ is a bounded sequence, and $N(x_\infty) \leq \sup_{n\in\mathbb{N}} N(x_n)$.

PROOF. By the reverse triangular inequality, $0 \leq |N(x_n) - N(x_\infty)| \leq N(x_n - x_\infty)$. Since $x_n \to x_\infty$, we have $N(x_n - x_\infty) \to 0$. By the squeeze theorem, $|N(x_n) - N(x_\infty)| \to 0$, meaning $N(x_n) \to N(x_\infty)$. The second part follows from the fact that any convergent sequence $(u_n)_{n\in\mathbb{N}}$ in $\mathbb{R}$ is bounded (i.e., $\sup_{n\in\mathbb{N}} |u_n| < +\infty$). Its limit satisfies $|u_\infty| \leq \sup_{n\in\mathbb{N}} |u_n|$. □

Proposition 5: Algebraic Operations on Limits

Let $(x_n)_{n\in\mathbb{N}}$ and $(y_n)_{n\in\mathbb{N}}$ be sequences in E, and let $(\lambda_n)_{n\in\mathbb{N}}$ be a sequence in $\mathbb{K}$. Assume:

1. $x_n \to x_\infty$ in E
2. $y_n \to y_\infty$ in E
3. $\lambda_n \to \lambda_\infty$ in $\mathbb{K}$

Then, $x_n + y_n \to x_\infty + y_\infty$ in E, and $\lambda_n y_n \to \lambda_\infty y_\infty$ in E.

PROOF. For $n \in \mathbb{N}$, set $z_n = x_n + \lambda_n y_n$ and $z_\infty = x_\infty + \lambda_\infty y_\infty$. We want to show $N(z_n - z_\infty) \to 0$. By the triangular inequality, $0 \leq N(z_n - z_\infty) \leq N(x_n - x_\infty) + N(\lambda_n y_n - \lambda_\infty y_\infty)$. By adding and subtracting $\lambda_\infty y_n$, we get $N(\lambda_n y_n - \lambda_\infty y_\infty) \leq |\lambda_n - \lambda_\infty| N(y_n) + |\lambda_\infty| N(y_n - y_\infty)$. Since $y_n \to y_\infty$, the sequence $(N(y_n))_{n\in\mathbb{N}}$ is bounded by some $M > 0$. Thus, $N(\lambda_n y_n - \lambda_\infty y_\infty) \leq M|\lambda_n - \lambda_\infty| + |\lambda_\infty| N(y_n - y_\infty)$. As $n \to \infty$, $|\lambda_n - \lambda_\infty| \to 0$ and $N(y_n - y_\infty) \to 0$, so $N(\lambda_n y_n - \lambda_\infty y_\infty) \to 0$. Consequently, $N(z_n - z_\infty) \to 0$, which completes the proof. □

Definition 6: Subsequence and Adherence Value

- A sequence $(y_n)_{n\in\mathbb{N}}$ is a **subsequence** (or extraction) of $(x_n)_{n\in\mathbb{N}}$ if there exists a strictly increasing function $\varphi : \mathbb{N} \to \mathbb{N}$ such that $y_n = x_{\varphi(n)}$.
- An element $l \in E$ is an **adherence value** (or accumulation point) of $(x_n)_{n\in\mathbb{N}}$ if there exists a strictly increasing extraction $\varphi : \mathbb{N} \to \mathbb{N}$ such that $x_{\varphi(n)} \to l$.

Problem 7: Accumulation Points of $\mathbb{Q}$

The set of rational numbers $\mathbb{Q}$ is countable, so it can be written as a sequence $\mathbb{Q} = \{q_0, q_1, \dots\}$. What are the accumulation points of the sequence $(q_n)_{n\in\mathbb{N}}$?

Remark 8

Because $\mathbb{Q}$ is dense in $\mathbb{R}$, every real number is an accumulation point. Thus, the set of accumulation points is all of $\mathbb{R}$.

3 Lecture 2: Open & Closed Subsets in Normed Vector Spaces

Let $(E, \|\cdot\|)$ be a fixed NVS over $\mathbb{K}$.

3.1 Open Subsets

Definition 1: Open Subset

A subset $U \subset E$ is **open** if for all $u \in U$, there exists $\varepsilon > 0$ such that $B(u, \varepsilon) \subset U$.

Example 2: Open Sets

- $\emptyset$ and E are open subsets.
- In $\mathbb{R}$, open intervals (a, b) are open subsets.

Proposition 3: Open Balls are Open Sets

In $(E, \|\cdot\|)$, open balls are open subsets.

Proof. Let $B = B(x_0, r) \subset E$ with $r > 0$. Let $y \in B$. We have $d(x_0, y) = \|x_0 - y\| = \delta < r$. Choose $\varepsilon_y = r - \delta = r - \|x_0 - y\| > 0$. We claim $B(y, \varepsilon_y) \subset B$. Let $z \in B(y, \varepsilon_y)$. By the triangular inequality: $d(x_0, z) \leq d(x_0, y) + d(y, z) = \|x_0 - y\| + \|y - z\| < \delta + \varepsilon_y = r$. Therefore, $z \in B$, which implies $B(y, \varepsilon_y) \subset B$. Thus, B is open. □

Proposition 4: Union of Open Balls

The open subsets of E are exactly the unions of open balls.

Proof. Let $U \subset E$ be an open subset. By definition, for every $u \in U$, there exists $\varepsilon_u > 0$ such that $B(u, \varepsilon_u) \subset U$. Then $U \subset \bigcup_{u \in U} B(u, \varepsilon_u) \subset U$. Thus, $U = \bigcup_{u \in U} B(u, \varepsilon_u)$, which is a union of open balls. □

3.2 Closed Subsets

Definition 5: Closed Subset

A subset $F \subset E$ is **closed** if its complement $F^c = E \setminus F$ is open.

Example 6: Closed Sets

$\emptyset$ and E are closed subsets of $(E, \|\cdot\|)$.

Proposition 7: Properties of Open and Closed Sets

1. Arbitrary unions of open subsets are open.
2. Finite intersections of open subsets are open.
3. Arbitrary intersections of closed subsets are closed.

4. Finite unions of closed subsets are closed.

Remark 8

An infinite intersection of open sets is not necessarily open. For example, in $\mathbb{R}$, the intersection $\bigcap_{n\geq 1}(-\frac{1}{n}, \frac{1}{n}) = \{0\}$ is not open.

PROOF. Let us prove the properties for open sets:

1. Let $(O_i)_{i\in I}$ be a family of open subsets. Let $\mathcal{O} = \bigcup_{i\in I} O_i$. If $x \in \mathcal{O}$, then there exists $i_0 \in I$ such that $x \in O_{i_0}$. Since O_{i_0} is open, there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subset O_{i_0} \subset \mathcal{O}$. Thus, $\mathcal{O}$ is open.

2. Assume I is finite and non-empty. Let $\mathcal{O} = \bigcap_{i\in I} O_i$. Let $u \in \mathcal{O}$. Then for all $i \in I$, $u \in O_i$. Since each O_i is open, there exists $\varepsilon_i > 0$ such that $B(u, \varepsilon_i) \subset O_i$. Because I is finite, $\varepsilon = \min_{i\in I} \varepsilon_i > 0$. Then $B(u, \varepsilon) \subset \bigcap_{i\in I} B(u, \varepsilon_i) \subset \mathcal{O}$. Thus, $\mathcal{O}$ is open.

The properties for closed sets follow symmetrically. □

3.3 Sequential Characterizations

Theorem 9: Sequential Characterization of Closed Subsets

For $F \subset E$, the following assertions are equivalent:

1. F is closed.

2. For every sequence $(x_n)_{n\in\mathbb{N}}$ in F converging in E, its limit $\lim x_n$ belongs to F.

PROOF. **$(1 \Rightarrow 2)$:** Assume F is closed. We proceed by contradiction. Assume there exists a sequence $(x_n)_{n\in\mathbb{N}}$ in F such that $x_n \to x_\infty \in E$, but $x_\infty \notin F$. Then $x_\infty \in F^c$. Since F is closed, F^c is open. Therefore, there exists $\varepsilon > 0$ such that $B(x_\infty, \varepsilon) \subset F^c$. Because $x_n \to x_\infty$, there exists N_ε such that for all $n \geq N_\varepsilon$, $x_n \in B(x_\infty, \varepsilon) \subset F^c$. This implies $x_n \in F \cap F^c = \emptyset$, which is a contradiction. Thus, $x_\infty \in F$.
$(\neg 1 \Rightarrow \neg 2)$: Assume F is not closed. Then F^c is not open. This means there exists $u^* \in F^c$ such that for all $\varepsilon > 0$, $B(u^*, \varepsilon) \not\subset F^c$, which is equivalent to saying $B(u^*, \varepsilon) \cap F \neq \emptyset$. For each $n \geq 1$, choose $x_n \in F \cap B(u^*, \frac{1}{n})$. This constructs a sequence $(x_n)_{n\in\mathbb{N}}$ in F. Since $d(x_n, u^*) < \frac{1}{n}$, we have $x_n \to u^*$ as $n \to \infty$. However, $u^* \notin F$, negating the second assertion. □

Theorem 10: Sequential Characterization of Open Subsets

For $U \subset E$, the following are equivalent:

1. U is open.

2. For every $u \in U$ and for every sequence $(x_n)_{n\in\mathbb{N}}$ converging to u in E, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $x_n \in U$.

3.4 Connectedness and Interior/Closure

Proposition 11: Clopen Subsets

If $X \subset E$ (or $\mathbb{R}$) is both closed and open, then $X = \emptyset$ or $X = E$.

PROOF. Assume for contradiction that $\emptyset \subsetneq X \subsetneq E$. Then there exists $x \in X$ and $y \in X^c$. Define $I = \{t \in [0,1] \mid x + t(y-x) \in X\}$. Since $x \in X$, $0 \in I$. Since $y \notin X$, $1 \notin I$. Let $z = \sup I$. Clearly, $z \leq 1$. By definition of supremum, there exists a sequence $(t_n)_{n\in\mathbb{N}}$ in I such that $t_n \to z$. The sequence $x_{t_n} = x + t_n(y-x)$ lies in X. Since X is closed, the limit $x_z = x + z(y-x)$ must belong to X. Thus, $z \in I$, which implies $z < 1$ (since $1 \notin I$). Because X is open and $x_z \in X$, there exists $\varepsilon > 0$ such that $B(x_z, \varepsilon) \subset X$. This means for sufficiently small $\delta > 0$, $x_{z+\delta} \in X$, implying $z + \delta \in I$. This contradicts $z = \sup I$. Therefore, $X = \emptyset$ or $X = E$. □

Definition 12: Interior and Closure

Let $X \subset E$.

- An element $x \in X$ is an **interior point** of X if there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subset X$.
- The **interior** of X, denoted $\mathring{X}$, is the set of all interior points of X. It is the largest open subset contained in X, given by $\mathring{X} = \bigcup_{\substack{O \subset X \\ O \text{ open}}} O$.
- An element $x \in E$ is **adherent** to X if for every $\varepsilon > 0$, $B(x, \varepsilon) \cap X \neq \emptyset$.
- The **topological closure** of X, denoted $\overline{X}$, is the set of all adherent points to X. It is the smallest closed subset containing X, given by $\overline{X} = \bigcap_{\substack{F \supset X \\ F \text{ closed}}} F$.

Proposition 13: Closure of a Ball

For $x_* \in E$ and $r > 0$, we have $\overline{B(x_*, r)} = \overline{B}(x_*, r)$, where $\overline{B(x_*, r)}$ is the topological closure of the open ball $B(x_*, r)$ and $\overline{B}(x_*, r)$ is the closed ball.

Remark 14

This is false in a general metric space. For example, let X be a set with at least two elements, and let d be the discrete metric $(d(x,y) = 1, \forall x \neq y)$. Then for all $x \in X$, $\overline{B(x,1)} = \{x\} \subsetneq X$.

PROOF. First, we show $\overline{B(x_*, r)} \subset \overline{B}(x_*, r)$. The closed ball $\overline{B}(x_*, r)$ is a closed set, because its complement $U = E \setminus \overline{B}(x_*, r)$ is open. To see this, let $y \in U$, so $\delta = d(x_*, y) > r$. Let $\varepsilon \in (0, \delta - r)$. By the reverse triangular inequality, $B(y, \varepsilon) \subset U$, so U is open. Since $\overline{B}(x_*, r)$ is closed and contains $B(x_*, r)$, it must contain its closure, so $\overline{B(x_*, r)} \subset \overline{B}(x_*, r)$.
Now we prove the opposite inclusion. Let $x \in \overline{B}(x_*, r)$. If $d(x_*, x) = N(x_* - x) < r$, then $x \in B(x_*, r) \subset \overline{B(x_*, r)}$. Otherwise, $d(x_*, x) = r$. We use the sequential characterization of the closure: $x \in \overline{X}$ if and only if there exists a sequence $(x_n)_{n\in\mathbb{N}}$ in X such that $x_n \to x$. For $n \geq 0$, set $x_n = x_* + \frac{n}{n+1}(x - x_*)$. Then $d(x_*, x_n) = N(\frac{n}{n+1}(x - x_*)) = \frac{n}{n+1} r < r$. So

$(x_n)_{n\in\mathbb{N}} \in B(x_*, r)^{\mathbb{N}}$ and $x_n \to x$. This implies $x \in \overline{B(x_*, r)}$. □

3.5 Dense Subsets

Definition 15: Dense Subset

Let $X, Y \subset E$ be subsets.

- X is dense in E if $\overline{X} = E$.
- X is dense in Y if $Y \subset \overline{X}$.

Example 16: Dense Sets

- $\overline{\mathbb{Z}} = \mathbb{Z}$, so $\mathbb{Z}$ is not dense in $\mathbb{R}$.
- $\overline{\mathbb{Q}} = \mathbb{R}$, so $\mathbb{Q}$ is dense in $\mathbb{R}$.
- Let $E = C^0([0,1], \mathbb{R})$ with the norm $||\cdot||_\infty$. Let P be the subspace of polynomials. By the Weierstrass approximation theorem, $\overline{P} = E$.

4 Lecture 3: Continuous Maps

Let (E, N_E) and (F, N_F) be $\mathbb{K}$-NVS. Let $\Omega \subset E$ be a non-empty open subset. Let $\varphi : \Omega \to F$.

Definition 1: Limit of a Map

Let $l \in F$. l is the limit of $\varphi(w)$ when $w \to w_0$ in Ω (with $w \neq w_0$) if for all $\varepsilon > 0$, there exists $\delta > 0$ such that for all $w \in \Omega$ with $0 < N_E(w - w_0) < \delta$, we have $\varphi(w) \in B_F(l, \varepsilon)$. Notation: $\lim_{w\to w_0} \varphi(w) = l$ or $\varphi(w) \to l$ as $w \to w_0$.

Proposition 2: Sequential Characterization of Limits

We have $\lim_{w\to w_0} \varphi(w) = l$ if and only if for every sequence $(w_n)_{n\in\mathbb{N}}$ in Ω such that $w_n \to w_0$, we have $\varphi(w_n) \to l$.

Definition 3: Continuity

- φ is continuous (C^0) at $w_0 \in \Omega$ if $\lim_{w\to w_0} \varphi(w) = \varphi(w_0)$.
- φ is continuous on Ω (notation: $\varphi \in C^0(\Omega, F)$) if φ is continuous at every $w \in \Omega$.

Example 4

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by $f(x, y) = 0$ if $(x, y) = (0, 0)$, and $f(x, y) = \frac{xy^2}{x^2+y^4}$ otherwise. It can be proven (as an exercise) that $f \notin C^0(\mathbb{R}^2, \mathbb{R})$.

Remark 5

Continuity of a map depends on the chosen norm.

Example 6: Norm Dependence

Let $E = C^0([0,1],\mathbb{R}) = F$. Let $N_E = ||\cdot||_\infty$ (sup norm) and $N_F = ||\cdot||_{L^1}$ (where $||f||_{L^1} = \int_0^1 |f(x)|dx$). Let $\Omega = E$, and let $\varphi : E \to F$ be the identity map, $f \mapsto f$. The map $\varphi : (E, ||\cdot||_\infty) \to (F, ||\cdot||_{L^1})$ is continuous. However, $\varphi^{-1} : (F, ||\cdot||_{L^1}) \to (E, ||\cdot||_\infty)$ is not continuous. To see this, consider $f_n(x) = x^n$ on $[0,1]$. $||f_n||_{L^1} = \frac{1}{n+1} \to 0$, so $f_n \to 0_F$ in $(F, ||\cdot||_{L^1})$. But $||f_n||_\infty = 1$ for all n, so f_n does not converge to 0_E in $(E, ||\cdot||_\infty)$.

Proposition 7: Operations on Continuous Functions

Let $f, g : \Omega \to F$ and $\lambda \in \mathbb{K}$.

- If f and g are continuous at w_0 (or on Ω), then $f + \lambda g$ is continuous at w_0 (or on Ω). Thus, $C^0(\Omega, F)$ is a $\mathbb{K}$-vector subspace of F^Ω.
- If $\mathbb{K} = F$ and f, g are continuous at w_0 (or on Ω), then $f \cdot g$ is continuous at w_0 (or on Ω).
- If $\mathbb{K} = F$, f is continuous at w_0, and $f(w_0) \neq 0$, then $\frac{1}{f}$ is defined on some $B(w_0, \varepsilon)$ and is continuous at w_0. (If $f \neq 0$ on Ω, then $\frac{1}{f}$ is continuous on Ω).

Proposition 8: Topological Characterization of Continuity

Let E, F be NVS and $\Omega \subset E$ open. Let $\varphi : \Omega \to F$. The following assertions are equivalent:

1. $\varphi \in C^0(\Omega, F)$.
2. For every open subset $U \subset F$, $\varphi^{-1}(U)$ is open (in E).

(Note: this also applies when $\Omega = E$).

Proof. $(1 \Rightarrow 2)$ Let $U \subset F$ be an open subset. Let $w \in \varphi^{-1}(U)$, so $\varphi(w) \in U$. Since U is open, there exists $\varepsilon > 0$ such that $B_F(\varphi(w), \varepsilon) \subset U$. Because φ is continuous at w, there exists $\delta > 0$ such that $\varphi(B_E(w, \delta) \cap \Omega) \subset B_F(\varphi(w), \varepsilon)$. Since Ω is open, there exists $\delta' \leq \delta$ such that $B_E(w, \delta') \subset \Omega$. Then $\varphi(B_E(w, \delta')) \subset B_F(\varphi(w), \varepsilon) \subset U$. Thus, $B_E(w, \delta') \subset \varphi^{-1}(U)$, which means $\varphi^{-1}(U)$ is open.
$(2 \Rightarrow 1)$ Let $w \in \Omega$ and $\varepsilon > 0$. Then $U = B_F(\varphi(w), \varepsilon)$ is open in F. By assumption, $\varphi^{-1}(U) = \varphi^{-1}(B_F(\varphi(w), \varepsilon))$ is open in E. Since $w \in \varphi^{-1}(U)$, there exists $\delta > 0$ such that $B_E(w, \delta) \subset \varphi^{-1}(U)$. This means for all $y \in B_E(w, \delta)$, $\varphi(y) \in B_F(\varphi(w), \varepsilon)$, i.e., $d_F(\varphi(w), \varphi(y)) < \varepsilon$. Thus, φ is continuous at w. □

4.1 Continuity of Linear Maps

Definition 9: Continuous Linear Maps

Let E, F be $\mathbb{K}$-NVS. Let $L(E,F) = \{L : E \to F \mid L \text{ is } \mathbb{K}\text{-linear}\}$. This is a $\mathbb{K}$-vector space. Let $\mathcal{L}(E,F) = \{L \in L(E,F) \mid L \text{ is continuous on } E\}$. This is a $\mathbb{K}$-vector subspace of $L(E,F)$.

Theorem 10: Characterization of Continuous Linear Maps

For $L \in L(E,F)$, the following assertions are equivalent:

1. L is continuous on E.
2. L is continuous at 0_E.
3. There exists $C \geq 0$ such that for all $x \in E$ with $N_E(x) = 1$, we have $N_F(L(x)) \leq C$.
4. There exists $C \geq 0$ such that for all $x \in E$, $N_F(L(x)) \leq CN_E(x)$.
5. $\sup_{x \neq 0} \frac{N_F(L(x))}{N_E(x)} < +\infty$.

PROOF. $1 \Rightarrow 2$: Obvious.
$2 \Rightarrow 3$: For $\varepsilon = 1$, since L is continuous at 0_E and $L(0_E) = 0_F$, there exists $\delta > 0$ such that for all $x \in B_E(0_E, \delta)$, we have $L(x) \in B_F(0_F, 1)$. Let $x \in E$ with $N_E(x) = 1$. Let $\tilde{x} = \frac{\delta}{2}x$. Then $N_E(\tilde{x}) = \frac{\delta}{2} < \delta$, so $\tilde{x} \in B_E(0_E, \delta)$. Thus, $L(\tilde{x}) \in B_F(0_F, 1)$, meaning $N_F(L(\frac{\delta}{2}x)) \leq 1$. By linearity, $\frac{\delta}{2}N_F(L(x)) \leq 1$, which implies $N_F(L(x)) \leq \frac{2}{\delta}$. We set $C = \frac{2}{\delta}$.
$3 \Rightarrow 4$: Let $x \in E \setminus \{0\}$. Then $\tilde{x} = \frac{x}{N_E(x)}$ has norm equal to 1. So $N_F(L(\tilde{x})) \leq C$. By linearity, $N_F\left(\frac{L(x)}{N_E(x)}\right) \leq C$, which implies $N_F(L(x)) \leq CN_E(x)$. (For $x = 0_E$, $N_F(L(0_E)) = 0 = C \cdot 0$).
$4 \Leftrightarrow 5$: By (4), the set $\{\frac{N_F(L(x))}{N_E(x)} \mid x \neq 0\}$ is bounded above by C, so its supremum is finite. Let this supremum be denoted as $\|L\|$. Then $\|L\| = \inf\{C \geq 0 \mid \forall x \in E, N_F(L(x)) \leq CN_E(x)\}$.
($5 \Rightarrow 1$): Assume $\sup_{x \neq 0} \frac{N_F(L(x))}{N_E(x)} = |||L||| < +\infty$. Then for all $x \in E$, $N_F(L(x)) \leq |||L|||N_E(x)$. For any $x_0 \in E$ and $x \in E$, by linearity we have $N_F(L(x) - L(x_0)) = N_F(L(x - x_0)) \leq |||L|||N_E(x - x_0)$. As $x \to x_0$, $N_E(x - x_0) \to 0$, which implies $N_F(L(x) - L(x_0)) \to 0$. Thus, L is continuous at any $x_0 \in E$, meaning L is continuous on E. This completes the cycle of equivalences. □

Proposition 11: Characterization of the Triple Norm

Let $\mathcal{N}_L = \left\{\frac{N_F(L(x))}{N_E(x)} \mid x \in E, x \neq 0\right\} \subset [0, +\infty)$. Then $\mathcal{N}_L = \{N_F(L(\tilde{x})) \mid \tilde{x} \in E, N_E(\tilde{x}) = 1\}$. Let $\mathcal{C} = \{C \geq 0 \mid \forall x \in E, N_F(L(x)) \leq CN_E(x)\}$ be the set of upper bounds. Then $\sup \mathcal{N}_L = \inf \mathcal{C}$.

4.2 Uniform Continuity and Lipschitz Continuity

Let $\varphi : \Omega \to F$.

Definition 12: Uniform Continuity

φ is **uniformly continuous** if: $\forall \varepsilon > 0, \exists \delta > 0$ such that $\forall x, x' \in \Omega$, if $d_E(x, x') < \delta$ then $d_F(\varphi(x), \varphi(x')) < \varepsilon$.

Definition 13: Lipschitz Continuity

φ is **Lipschitz** of parameter $\alpha \geq 0$ (or α-Lipschitz) if: $\forall x, x' \in \Omega, d_F(\varphi(x), \varphi(x')) \leq \alpha d_E(x, x')$.

Example 14: Lipschitz Maps

1. By the reverse triangular inequality, $\forall x, x' \in E, |N(x) - N(x')| \leq N(x - x')$. Thus, $N : E \to \mathbb{R}$ is 1-Lipschitz.
2. Let $L \in \mathcal{L}(E, F)$. Then $|||L||| = \sup_{x \neq 0} \frac{N_F(L(x))}{N_E(x)} < +\infty$. For all $x, x' \in E$: $d_F(L(x), L(x')) = N_F(L(x) - L(x')) = N_F(L(x - x')) \leq |||L|||N_E(x - x')$. Thus, L is $|||L|||$-Lipschitz.

Proposition 15: Implications of Continuity Types

For $\varphi : \Omega \to F$: φ is Lipschitz $\Rightarrow$ φ is uniformly continuous (on Ω) $\Rightarrow$ $\varphi \in C^0(\Omega, F)$.

Remark 16

These implications are strict.

- $f : [0, +\infty) \to \mathbb{R}$, $x \mapsto \sqrt{x}$ is uniformly continuous on $[0, +\infty)$ but not Lipschitz.
- $f : \mathbb{R} \to \mathbb{R}$, $x \mapsto x^2$ is continuous on $\mathbb{R}$ but not uniformly continuous on $\mathbb{R}$.

4.3 Properties of the Triple Norm

Definition 17: Triple Norm

For $L \in \mathcal{L}(E, F)$, the triple norm (or operator norm) of L induced by N_E and N_F is: $|||L||| = \sup_{x \neq 0} \frac{N_F(L(x))}{N_E(x)} \in [0, +\infty)$.

Proposition 18: Properties of Triple Norm

1. For $L \in \mathcal{L}(E, F)$ and $\forall x \in E$, $N_F(L(x)) \leq |||L|||N_E(x)$.
2. $||| \cdot |||$ is a norm on $\mathcal{L}(E, F)$.

Proof. Let us prove that $||| \cdot |||$ is a norm on $\mathcal{L}(E, F)$.

- **Positive definiteness:** Assume $|||L||| = 0$. Then for all $x \in E$, $N_F(L(x)) = 0$, which implies $L(x) = 0_F$. Thus $L = 0_{\mathcal{L}(E,F)}$.
- **Absolute homogeneity:** Let $L \in \mathcal{L}(E, F)$ and $\lambda \in \mathbb{K}$. $|||\lambda L||| = \sup_{x \neq 0} \frac{N_F((\lambda L)(x))}{N_E(x)} = \sup_{x \neq 0} \frac{|\lambda| N_F(L(x))}{N_E(x)} = |\lambda| \sup_{x \neq 0} \frac{N_F(L(x))}{N_E(x)} = |\lambda||||L|||$.
- **Triangular inequality:** Let $x \in E$ and $L, \tilde{L} \in \mathcal{L}(E, F)$. $N_F((L + \tilde{L})(x)) = N_F(L(x) + \tilde{L}(x)) \leq N_F(L(x)) + N_F(\tilde{L}(x)) \leq |||L|||N_E(x) + |||\tilde{L}|||N_E(x) = (|||L||| + |||\tilde{L}|||)N_E(x)$. Thus, $|||L||| + |||\tilde{L}|||$ is an upper bound for $\frac{N_F((L+\tilde{L})(x))}{N_E(x)}$, which implies $|||L + \tilde{L}||| \leq |||L||| + |||\tilde{L}|||$.

□

Proposition 19: Unbounded Linear Maps

$L \in L(E,F)$ is not bounded on $B_E(0,1)$ if and only if $L \notin \mathcal{L}(E,F)$ if and only if there exists a sequence $(x_n)_{n\in\mathbb{N}}$ in E such that $N_E(x_n) \to 1$ and $N_F(L(x_n)) \to +\infty$.

PROOF. Left as an exercise. □

Proposition 20: Computing the Operator Norm

For $L \in \mathcal{L}(E,F)$ and $C \geq 0$:

1. If for all $x \in E$, $N_F(L(x)) \leq CN_E(x)$, then $|||L||| \leq C$.
2. If there exists a sequence $(x_n)_{n\in\mathbb{N}}$ in E such that $N_E(x_n) \to 1$ and $N_F(L(x_n)) \to C$, then $|||L||| \geq C$.

(If both conditions are satisfied, then $|||L||| = C$).

Example 21: Operator on l^1

Let $l^1(\mathbb{N}) = \{\underline{x} = (x_n)_{n\in\mathbb{N}} \in \mathbb{R}^{\mathbb{N}} \mid \|\underline{x}\|_{l^1} < +\infty\}$, where $\|\underline{x}\|_{l^1} = \sum_{n\geq 0} |x_n|$. Let $\lambda = (\lambda_n)_{n\in\mathbb{N}} \in l^\infty(\mathbb{N})$, so $\|\lambda\|_\infty = \sup_n |\lambda_n| < +\infty$. Define $\Lambda : l^1(\mathbb{N}) \to l^1(\mathbb{N})$ by $\Lambda((x_n)_{n\in\mathbb{N}}) = (\lambda_n x_n)_{n\in\mathbb{N}}$. We have: $\|\Lambda(\underline{x})\|_{l^1} = \sum_{n\geq 0} |\lambda_n x_n| \leq \|\lambda\|_\infty \sum_{n\geq 0} |x_n| = \|\lambda\|_\infty \|\underline{x}\|_{l^1}$. This implies $\Lambda(\underline{x}) \in l^1(\mathbb{N})$, so Λ is well-defined. Λ is clearly $\mathbb{R}$-linear, so $\Lambda \in L(l^1(\mathbb{N}), l^1(\mathbb{N}))$. From the inequality above, we get that $\Lambda \in \mathcal{L}(l^1(\mathbb{N}), l^1(\mathbb{N}))$ and $|||\Lambda||| \leq \|\lambda\|_\infty$. (Exercise: prove that $|||\Lambda||| = \|\lambda\|_\infty$ using the second fact from the proposition above).

5 Lecture 4: Compactness

Let (E,N) be a NVS over $\mathbb{K}$.

Definition 1: Compactness

A subset $K \subset E$ is **compact** if $K \neq \emptyset$ and for every sequence $(x_n)_{n\in\mathbb{N}}$ in K, there exists a strictly increasing extraction $\varphi : \mathbb{N} \to \mathbb{N}$ such that the subsequence $(x_{\varphi(n)})_{n\in\mathbb{N}}$ converges in K.

Example 2: Compact Sets

- In $E = \mathbb{R}$, let $I = [a,b]$ with $-\infty < a \leq b < +\infty$. Then I is compact.
- By the Bolzano-Weierstrass theorem, any bounded sequence in $\mathbb{R}$ admits a convergent subsequence. The proof involves defining $S_n = \inf_{k\geq n} x_k$.
- The rectangle $R = [a,b] \times [a',b']$ is compact in $\mathbb{R}^2$ (for any norm, e.g., $\|\cdot\|_\infty$). (Exercise).

Proposition 3: Compact Implies Closed and Bounded

For $K \subset E$, if K is compact, then K is bounded and closed.

PROOF. We prove this by contrapositive: if K is not bounded or K is not closed, then K is not compact.
Assume K is not bounded. Then for all $n \in \mathbb{N}$, $K \not\subset B(0,n)$. This means there exists $x_n \in K \setminus B(0,n)$, so $N_E(x_n) \geq n$. Because any convergent sequence in E is bounded, the sequence $(x_n)_{n\in\mathbb{N}}$ (where $N_E(x_n) \geq n$) can never contain a convergent subsequence. Thus, K is not compact.
Assume K is not closed. By the sequential criterion for closedness, there exists a sequence $(x_n)_{n\in\mathbb{N}}$ in K such that $x_n \to x_\infty \in E$, but $x_\infty \notin K$. Then any subsequence $(x_{\varphi(n)})_{n\in\mathbb{N}}$ also converges to x_∞. Since $x_\infty \notin K$, the subsequence does not converge in K. Thus, K is not compact. □

Proposition 4: Closed Subsets of Compact Sets

Every closed subset of a compact set is compact.

PROOF. Let K be a compact subset of a normed vector space (E,N), and let $F \subseteq K$ be closed in E.
Consider any sequence $(x_n)_{n\in\mathbb{N}}$ in F. Since K is compact, there exists a subsequence $(x_{\varphi(n)})_{n\in\mathbb{N}}$ converging to some $l \in K$. As F is closed and $x_{\varphi(n)} \in F$ for every n, we have $l \in F$. Thus F is compact. □

Remark 5

Warning: The converse (bounded and closed implies compact) is generally false in infinite-dimensional spaces.

Example 6: Closed and Bounded but Not Compact

Let $l^\infty(\mathbb{N}) = \{\underline{x} = (x_n)_{n\in\mathbb{N}} \in \mathbb{R}^{\mathbb{N}} \mid \|\underline{x}\|_\infty = \sup_{n\in\mathbb{N}} |x_n| < +\infty\}$. Let $S = \{\underline{x} \in l^\infty(\mathbb{N}) \mid N(\underline{x}) = 1\} = N^{-1}(\{1\})$. S is clearly bounded. Since $\{1\}$ is closed in $\mathbb{R}$ and N is 1-Lipschitz (hence continuous), $S = N^{-1}(\{1\})$ is closed in $l^\infty(\mathbb{N})$. Thus, S is bounded and closed. However, consider the sequence $(e_n)_{n\in\mathbb{N}}$ where $e_n = (\delta_{n,k})_{k\in\mathbb{N}} = (0,\dots,0,1,0,\dots)$ (the 1 is at the n-th position). For all n, $\|e_n\|_\infty = 1$, so $e_n \in S$. For $n \neq m$, $\|e_n - e_m\|_\infty = 1$. Therefore, no subsequence of $(e_n)_{n\in\mathbb{N}}$ can be Cauchy, and hence no subsequence can converge. Thus, S is not compact.

Theorem 7: Riesz's Theorem

In a NVS (E,N), the closed unit ball $\overline{B_E}(0,1)$ is compact if and only if $\dim E < +\infty$.

Proposition 8: Images of Compact Sets

If $\varphi : \Omega \to F$ is continuous on Ω and $K \subset \Omega$ is compact, then $\varphi(K)$ is compact.

PROOF. Assume $K \neq \emptyset$. We set $\mathcal{K} = \varphi(K)$. Let $(y_n)_{n\in\mathbb{N}} \in \mathcal{K}^{\mathbb{N}}$. We know that for all $n \in \mathbb{N}$, there exists $x_n \in K$ such that $y_n = \varphi(x_n)$. Since K is compact and $(x_n)_{n\in\mathbb{N}} \in K^{\mathbb{N}}$, there exists a strictly increasing extraction $\psi : \mathbb{N} \to \mathbb{N}$ such that $x_{\psi(n)} \to x_\infty$ for some $x_\infty \in K$. Because φ is continuous at $x_\infty \in K \subset \Omega$, we have that $\varphi(x_{\psi(n)}) \to \varphi(x_\infty)$. This means $y_{\psi(n)} \to y_\infty \in \mathcal{K}$, so $\mathcal{K}$ is compact. □

Corollary 9: Extreme Value Theorem

If $K \subset E$ is compact and $f : K \to \mathbb{R}$ is continuous, then f attains its extrema on K.

PROOF. By the previous proposition, $\mathcal{K} = f(K) \subset \mathbb{R}$ is compact. Therefore, $\mathcal{K}$ is bounded and closed. Since $\mathcal{K}$ is bounded from above, $\alpha = \sup \mathcal{K} < +\infty$. Since $\mathcal{K}$ is closed, $\alpha \in \mathcal{K}$. Thus, there exists $x^* \in K$ such that $f(x^*) = \alpha$. For every $x \in K$, we have $f(x) \leq \alpha = f(x^*)$, i.e., $f(x^*) = \sup_{x \in K} f(x)$. The proof for the infimum is similar. □

6 Lecture 5: Compactness II

Let $E = \mathbb{R}^d$ (for $d \geq 0$) or $\mathbb{C}^d$. We equip E with the sup norm $\|x\|_\infty = \max_{i=1,\dots,d} |x^i|$.

Proposition 1: Component-wise Convergence

Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in E, with $x_n = (x_n^1, \dots, x_n^d)$ for all $n \in \mathbb{N} \cup \{\infty\}$. Then $x_n \to x_\infty$ in $(E, \|\cdot\|_\infty)$ if and only if for all $i = 1, \dots, d$, $x_n^i \to x_\infty^i$ in $\mathbb{K}$.

Proposition 2: Heine-Borel Theorem in $\mathbb{R}^d$

For $K \subseteq E = \mathbb{R}^d$, K is compact if and only if K is bounded and closed.

PROOF. The fact that compact implies bounded and closed was proved in Lecture 4. Now assume $K \neq \emptyset$, bounded, and closed. Since K is bounded, there exists $\rho > 0$ such that $K \subseteq B(0, \rho)$ for $\|\cdot\|_\infty$. This means $K \subseteq [-\rho, \rho]^d \subset \mathbb{R}^d$. Because closed subsets of a compact set are compact, it is sufficient to prove that $[-\rho, \rho]^d$ is compact. We prove this by induction on $d \geq 0$.

- $d = 0$: Obvious.
- $d = 1$: True by the Bolzano-Weierstrass theorem.
- $d \geq 2$: We have a natural linear isomorphism $\mathbb{R}^d \to \mathbb{R}^{d-1} \times \mathbb{R}$, sending $(x^1, \dots, x^d)$ to $((x^1, \dots, x^{d-1}), x^d)$. Under this isomorphism, the norm corresponds to $\max(\|\cdot\|_{\mathbb{R}^{d-1},\infty}, |\cdot|_{\mathbb{R}})$. Then $[-\rho, \rho]^d = [-\rho, \rho]^{d-1} \times [-\rho, \rho]$. The first factor is compact by the induction hypothesis, and the second is compact in $(\mathbb{R}, |\cdot|)$. By the following lemma, the product of two compact sets is compact.

□

Lemma 3: Product of Metric Spaces

Let (X_1, d_1) and (X_2, d_2) be two metric spaces. Let $X = X_1 \times X_2$, and define $d : X^2 \to \mathbb{R}^+$ by $d((x_1, x_2), (y_1, y_2)) = \max(d_1(x_1, y_1), d_2(x_2, y_2))$.

1. d is a metric on X.
2. If $K_i \subset X_i$ are non-empty compact subsets for $i = 1, 2$, then $K = K_1 \times K_2$ is compact in X for d.

PROOF. Left as an exercise. □

6.1 Equivalence of Norms

Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, and let E be a NVS over $\mathbb{K}$.

Definition 4: Weaker Norm

Let N_1, N_2 be two norms on E. We say N_2 is **weaker** than N_1 if there exists $C > 0$ such that $N_2 \leq CN_1$, meaning $\forall x \in E, N_2(x) \leq CN_1(x)$.

Example 5

- On $\mathbb{R}^2$, $N_1 = \|\cdot\|_\infty$ and $N_2 = \|\cdot\|_2$. For $x, y \in \mathbb{R}^2$, $\|(x,y)\|_2 = \sqrt{x^2+y^2} \leq \sqrt{2\max(x^2,y^2)} = \sqrt{2}\|(x,y)\|_\infty$. Thus, $\|\cdot\|_2$ is weaker than $\|\cdot\|_\infty$.
- Let $E = C^1([0,1],\mathbb{R})$. Consider the norms $N_1(f) = \|f\|_\infty + \|f'\|_\infty$ and $N_2(f) = \|f\|_\infty$. Clearly, $N_2(f) \leq 1 \cdot N_1(f)$, so N_2 is weaker than N_1. However, N_1 is not weaker than N_2. Consider $f_n(x) = x^n$. We have $\|f_n\|_\infty = 1$, but $N_1(f_n) = 1 + n$. There is no constant $C \geq 0$ such that $1 + n \leq C \cdot 1$ for all n, so N_1 is not weaker than N_2.

Proposition 6: Properties of Weaker Norms

Let N_1, N_2 be norms on E.

1. If N_2 is weaker than N_1, then for any sequence $(x_n)_{n\in\mathbb{N}}$ in E:
   - If $(x_n)_{n\in\mathbb{N}}$ is bounded for N_1, it is bounded for N_2.
   - If $(x_n)_{n\in\mathbb{N}}$ converges for N_1, it converges for N_2.
2. If N_2 is not weaker than N_1, then there exists a sequence $(x_n)_{n\in\mathbb{N}}$ in E that is bounded for N_1 but not for N_2, and similarly for convergence.

Definition 7: Equivalent Norms

N_1 and N_2 are **equivalent** (notation: $N_1 \sim N_2$) if N_1 is weaker than N_2 and N_2 is weaker than N_1. This is equivalent to: $\exists c_1, c_2 > 0$ such that $\forall x \in E$, $c_1 N_1(x) \leq N_2(x) \leq c_2 N_1(x)$.

Proposition 8

Equivalence $\sim$ is an equivalence relation on the set of all norms on E.

Example 9

On $\mathbb{R}^2$, $\|\cdot\|_\infty \leq \|\cdot\|_2 \leq \sqrt{2}\|\cdot\|_\infty$, so $\|\cdot\|_\infty \sim \|\cdot\|_2$.

Theorem 10: Equivalence of Norms in Finite Dimensions

On any finite-dimensional vector space E, all norms are equivalent.

Lemma 11: Reduction to Real Scalars

Let E be a $\mathbb{C}$-vector space, and N a norm on E.

1. Let $E|_{\mathbb{R}}$ denote E viewed as an $\mathbb{R}$-vector space (restriction of scalars). Then N is a norm on $E|_{\mathbb{R}}$.
2. If N_1, N_2 are norms on E, then $N_1 \sim N_2$ as norms on E if and only if $N_1 \sim N_2$ as norms on $E|_{\mathbb{R}}$.

Proof of Theorem 10 for $E = \mathbb{R}^d$. Assume $d > 0$. Let N be a norm on E. Let us prove $N \sim \|\cdot\|_\infty$. For $x \in E$, we can write $x = \sum_{i=1}^d x^i e_i$ using the canonical basis $(e_i)_{i=1}^d$. By the triangular inequality, $N(x) = N(\sum_{i=1}^d x^i e_i) \leq \sum_{i=1}^d |x^i| N(e_i)$. Let $C = \sum_{i=1}^d N(e_i) > 0$. Then $N(x) \leq \|x\|_\infty C$, so N is weaker than $\|\cdot\|_\infty$.
Now we prove the opposite. Let $S = \{x \in \mathbb{R}^d \mid \|x\|_\infty = 1\}$. Because $\|\cdot\|_\infty : (\mathbb{R}^d, \|\cdot\|_\infty) \to \mathbb{R}$ is 1-Lipschitz and therefore continuous, $S = \|\cdot\|_\infty^{-1}(\{1\})$ is closed. Since S is also bounded, S is compact in $(\mathbb{R}^d, \|\cdot\|_\infty)$. Since N is weaker than $\|\cdot\|_\infty$, we have $|N(x) - N(y)| \leq N(x-y) \leq C\|x-y\|_\infty$. Thus, $N : (\mathbb{R}^d, \|\cdot\|_\infty) \to \mathbb{R}$ is Lipschitz, and therefore continuous. Since S is compact and N is continuous, N attains its infimum on S. So there exists $x^* \in S$ such that for all $x \in S$, $N(x^*) \leq N(x)$. Since $x^* \in S$, $x^* \neq 0$, which implies $\nu = N(x^*) > 0$. For any $y \neq 0 \in E$, $\frac{y}{\|y\|_\infty} \in S$, so $\nu \leq N\left(\frac{y}{\|y\|_\infty}\right)$. By linearity, $\|y\|_\infty \leq \frac{1}{\nu} N(y)$, meaning $\|\cdot\|_\infty$ is weaker than N. Thus, $N \sim \|\cdot\|_\infty$. □

Corollary 12: General Finite Dimensional Case

Let E be a finite-dimensional vector space over $\mathbb{K}$ ($\mathbb{R}$ or $\mathbb{C}$). All norms are equivalent on E.

Proof. Let N_1, N_2 be norms on E, and let $d = \dim E > 0$. Choose a basis $(e_i)_{i=1}^d$ for E. The map $\psi : \mathbb{K}^d \to E$ defined by $\psi((x_i)_{i=1}^d) = \sum_{i=1}^d x_i e_i$ is a vector space isomorphism. Define norms on $\mathbb{K}^d$ by $\tilde{N}_j = N_j \circ \psi$ for $j = 1, 2$. By the previous theorem, $\tilde{N}_1 \sim \tilde{N}_2$ on $\mathbb{K}^d$. Because ψ is an isomorphism, this implies $N_1 \sim N_2$ on E. □

Corollary 13: Topological Properties in Finite Dimensions

On a finite-dimensional NVS, there is a unique notion of:

- open / closed / bounded / compact subsets.
- convergence of sequences.
- continuity of maps.

In finite dimensions, there is no need to specify the norm for these topological properties.

Corollary 14: Continuity in Finite Dimensions

Let E be a finite-dimensional normed vector space and F be any normed vector space. Then every linear map from E to F is continuous, i.e., $L(E, F) = \mathcal{L}(E, F)$.

Proof. Assume $d = \dim_K E > 0$ (the $d = 0$ case is trivial). Let $(e_i)_{i=1}^d$ be a K-basis of E. For any $x \in E$, we can write $x = \sum_{i=1}^d x_i e_i$, where $x_i \in K$ for all i. For a linear map $L \in$

$L(E,F)$, we have: $N_F(L(x)) = N_F\left(L\left(\sum_{i=1}^d x_i e_i\right)\right) = N_F\left(\sum_{i=1}^d x_i L(e_i)\right)$. By the triangular inequality and absolute homogeneity: $N_F(L(x)) \leq \sum_{i=1}^d N_F(x_i L(e_i)) = \sum_{i=1}^d |x_i| N_F(L(e_i))$. Let $||x||_\infty = \max_i |x_i|$ and $C = \sum_{i=1}^d N_F(L(e_i))$. Then $N_F(L(x)) \leq ||x||_\infty \cdot C$. Thus, L is bounded with respect to the $||\cdot||_\infty$ norm. Since all norms are equivalent in finite dimensions, L is continuous, meaning $L(E,F) = \mathcal{L}(E,F)$. □

Theorem 15: Heine-Cantor Theorem

Let K be a compact subset of a normed vector space E, and let $f : K \to F$ be a continuous function. Then f is uniformly continuous on K.

Proof. Assume for contradiction that $f|_K$ is not uniformly continuous. Then there exists $\epsilon > 0$ such that for all $\delta > 0$, there exist points $(\tilde{x}_\delta, \tilde{y}_\delta) \in K^2$ satisfying $d_E(\tilde{x}_\delta, \tilde{y}_\delta) < \delta$ but $d_F(f(\tilde{x}_\delta), f(\tilde{y}_\delta)) \geq \epsilon$. For each $n \in \mathbb{N}$, we set $\delta = \frac{1}{n+1}$, and define $x_n = \tilde{x}_{1/(n+1)} \in K$ and $y_n = \tilde{y}_{1/(n+1)} \in K$. Since K is compact, the sequence $(x_n)_{n\in\mathbb{N}}$ has a convergent subsequence. Thus, there exists a strictly increasing function $\varphi : \mathbb{N} \to \mathbb{N}$ such that $x_{\varphi(n)} \to x_\infty \in K$. We know $d_E(x_{\varphi(n)}, y_{\varphi(n)}) \leq \frac{1}{\varphi(n)+1} \leq \frac{1}{n+1} \to 0$. Thus, $y_{\varphi(n)}$ also converges to x_∞. Because f is continuous at x_∞, we have $f(x_{\varphi(n)}) \to f(x_\infty)$ and $f(y_{\varphi(n)}) \to f(x_\infty)$. This implies that $d_F(f(x_{\varphi(n)}), f(y_{\varphi(n)})) \to 0$ as $n \to \infty$. This contradicts our assumption that the distance is always $\geq \epsilon$. □

7 Lecture 6: Completeness

Let (E,N) be a NVS over K (where $K = \mathbb{R}$ or $\mathbb{C}$).

7.1 Series and Banach Spaces

Remark 1

A series $\sum_n x_n$ in E is formally a pair of sequences $((x_n)_{n\in\mathbb{N}}, (S_N)_{N\in\mathbb{N}})$, where $S_N = \sum_{k=0}^N x_k$ is the sequence of partial sums.

Definition 2: Convergence of Series

- A series $\sum x_n$ in E **converges** if the sequence of partial sums $(S_N)_{N\in\mathbb{N}}$ converges in E. In this case, $\sum_{n=0}^\infty x_n = \lim_{N\to\infty} S_N$ is the sum of the series.
- A series $\sum x_n$ **normally converges** if the real series $\sum_{k=0}^\infty N(x_k)$ converges in $\mathbb{R}$.

Definition 3: Completeness (Banach Space)

A normed vector space (E,N) is **complete** (or is a **Banach space**) if for every series $\sum x_n$ in E, normal convergence implies convergence.

Example 4: Incomplete Space

Let $E = \{\underline{x} = (x_n)_{n\in\mathbb{N}} \in \mathbb{R}^{\mathbb{N}} \mid \text{supp}(x) \text{ is finite}\} \simeq \bigoplus_{n\in\mathbb{N}} \mathbb{R}$, equipped with the sup norm $\|\underline{x}\|_\infty = \sup_{n\in\mathbb{N}} |x_n| < \infty$. For $n \geq 1$, define $x_n = (0, \ldots, \frac{1}{n^2}, 0, \ldots)$ where the non-zero entry is at index n. Then $\|x_n\|_\infty = \frac{1}{n^2}$. The series $\sum x_n$ normally converges because

$\sum_{n=1}^{\infty} \frac{1}{n^2} < +\infty$. However, the partial sum $S_N = \sum_{k=1}^{N} x_k = (1, \frac{1}{4}, \frac{1}{9}, \dots, \frac{1}{N^2}, 0, \dots)$. Assume for contradiction that $S_N \to x_\infty \in E$. Since x_∞ has finite support, there exists n_0 such that for all $n > n_0$, the n-th coordinate of x_∞ is 0. For every $n > n_0 \geq 1$, the n-th coordinate of S_N (for $N \geq n$) is $\frac{1}{n^2}$, meaning $\|x_\infty - S_N\|_\infty \geq \frac{1}{n^2}$. Thus, $(S_N)_{N \in \mathbb{N}}$ does not converge in E, meaning E is not complete.

Example 5: Complete Space

$E = C^0([0,1], K)$ endowed with $\|\cdot\|_\infty$ is complete. Let $\sum f_n$ normally converge in E. Let $f = \sum_{n=0}^{\infty} \|f_n\|_\infty < +\infty$. For any fixed $x \in [0,1]$, $\sum_{k=0}^{N} |f_k(x)| \leq \sum_{k=0}^{N} \|f_k\|_\infty \leq f < +\infty$. Thus, the real series $\sum f_n(x)$ normally converges in $\mathbb{R}$ (or $\mathbb{C}$). Since $\mathbb{R}$ is complete, we can define a point-wise limit function $g : [0,1] \to \mathbb{R}$ as $g(x) = \sum_{n=0}^{\infty} f_n(x)$. We have $|g(x) - \sum_{k=0}^{N} f_k(x)| = |\sum_{n=N+1}^{\infty} f_n(x)| \leq \sum_{n=N+1}^{\infty} \|f_n\|_\infty$. Taking the supremum over $x \in [0,1]$, we obtain $\|g - S_N\|_\infty \leq \sum_{n=N+1}^{\infty} \|f_n\|_\infty \to 0$ as $N \to \infty$. This shows $S_N \to g$ uniformly on $[0,1]$. Because each S_N is continuous, the uniform limit g is also continuous, so $g \in E$ and $g = \sum_{n=0}^{\infty} f_n$.

Theorem 6: Completeness of Finite-Dimensional Spaces

For $d \geq 0$, $E = K^d$ is complete for any norm.

Corollary 7

Any finite-dimensional normed vector space is complete.

Lemma 8: Equivalence and Completeness

1. Let N and $\tilde{N}$ be equivalent norms on E. A series normally converges for N if and only if it normally converges for $\tilde{N}$. Thus, (E, N) is complete if and only if $(E, \tilde{N})$ is complete.
2. When $K = \mathbb{C}$, E is a $\mathbb{C}$-vector space. (E, N) is complete over $\mathbb{C}$ if and only if it is complete over $\mathbb{R}$.

Proof of Theorem 6. From the facts above, it is sufficient to prove the theorem for $(\mathbb{R}^d, \|\cdot\|_\infty)$. For $d = 1$, $(\mathbb{R}, |\cdot|)$ is complete (known from Semester 1 analysis). For $d \geq 2$, let $\sum x_n$ normally converge in $\mathbb{R}^d$, where $x_n = (x_n^1, \dots, x_n^d) \in \mathbb{R}^d$. Since $|x_n^i| \leq \|x_n\|_\infty$ for all $1 \leq i \leq d$, the series $\sum x_n^i$ normally converges in $\mathbb{R}$. Because $\mathbb{R}$ is complete, each coordinate sequence converges, which implies $\sum x_n$ converges in $\mathbb{R}^d$. □

7.2 Cauchy Completeness

Definition 9: Cauchy Sequence

Let (X, d) be a metric space. A sequence $(x_n)_{n \in \mathbb{N}}$ is **Cauchy** if for all $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $m, n \geq N$, $d(x_n, x_m) < \epsilon$.

Definition 10: Cauchy Complete Metric Space

A metric space (X, d) is **Cauchy complete** if every Cauchy sequence $(x_n)_{n\in\mathbb{N}} \in X^{\mathbb{N}}$ converges.

Theorem 11: Equivalence of Completeness Definitions

Let (E, N) be a normed vector space, and d_N be the induced distance. Then (E, N) is complete (using the series definition) if and only if (E, d_N) is Cauchy complete.

7.3 Fixed Point Theorems

Theorem 12: Banach-Picard Fixed Point Theorem

Let (E, N) be a complete NVS, and let $C \subset E$ be a non-empty closed subset. Let $f : C \to C$ be a contracting map, meaning there exists $k \in [0, 1)$ such that for all $x, y \in C$, $d(f(x), f(y)) \leq kd(x, y)$. Then there exists a unique $x \in C$ that is a fixed point for f (i.e., $f(x) = x$).

Proof. **Uniqueness:** Assume there are two fixed points $x_1, x_2 \in C$. Then $d(x_1, x_2) = d(f(x_1), f(x_2)) \leq kd(x_1, x_2)$. Since $k < 1$, this implies $d(x_1, x_2) = 0$, so $x_1 = x_2$. Thus, the fixed point is at most a singleton.
Existence: Pick an arbitrary $x_0 \in C$. We define a sequence inductively by $x_{n+1} = f(x_n) \in C$ for all $n \in \mathbb{N}$. Let $u_n = x_{n+1} - x_n$. For $n \geq 1$, we have: $N(u_n) = N(x_{n+1} - x_n) = N(f(x_n) - f(x_{n-1})) = d(f(x_n), f(x_{n-1}))$. Since f is a contraction, $N(u_n) \leq kd(x_n, x_{n-1}) = kN(x_n - x_{n-1}) = kN(u_{n-1})$. By induction, $N(u_n) \leq k^n N(u_0)$. Then $\sum_{n=0}^{N} N(u_n) \leq \sum_{n=0}^{N} k^n N(u_0) \leq \frac{N(u_0)}{1-k} < +\infty$. Therefore, the series $\sum u_n$ normally converges in E. Because E is complete, the series converges in E. Since $\sum_{n=0}^{N} u_n = x_{N+1} - x_0$, the sequence $x_n = x_0 + \sum_{k=0}^{n-1} u_k$ converges to some $x_\infty \in E$. Because C is closed and $x_n \in C$ for all n, we have $x_\infty \in C$. Finally, since f is Lipschitz, it is continuous on C. Because $x_n \to x_\infty$, we get $f(x_n) \to f(x_\infty)$. But $f(x_n) = x_{n+1}$, and $x_{n+1} \to x_\infty$, so $f(x_\infty) = x_\infty$. □